

Control work 5, and work on the mistakes

The definite integral in one paper — where the limits and the sketch carry the marks.

LESSON 58–59

2 × 40 MIN

ALGEBRA 11, NAZORAT ISHI 5

P1 · CHAPTER 10 REVIEW

LESSON CLOCK

3

40'

10'

25'

12'

Instructions

3 min

The paper

40 min

Answers

10 min

Rewrite

25 min

The limits drill

12 min

LEARNING OBJECTIVES

- Apply the definite-integral methods under time.
- Find the limits of integration from the geometry, unprompted.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National

pp. 269–272

Cambridge

Pure Mathematics 1, pp. 241–244

Workbook

Control paper E

01

Explanation

The paper — 25 marks, 40 minutes

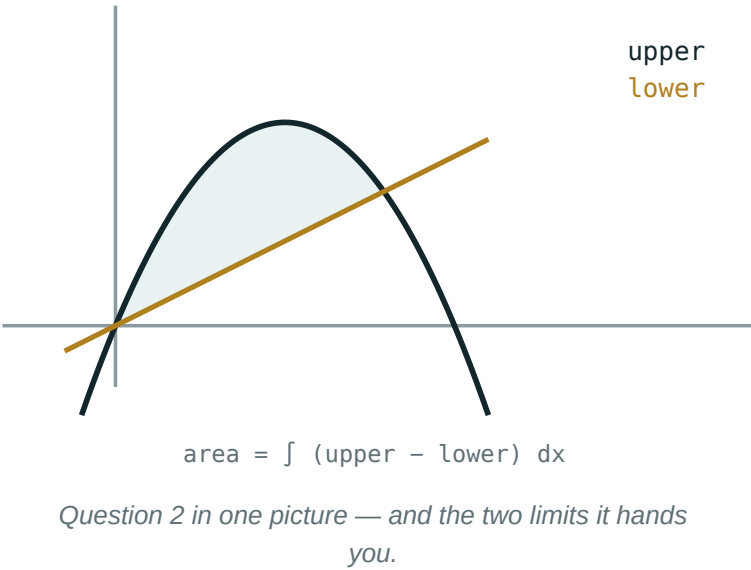
Q	TASK	MARKS	FROM
1	Evaluate $\int_1^4 (x^2 - 2x) \, dx$ and $\int_1^9 \frac{1}{\sqrt{x}} \, dx$	5	L49–51
2	Find the area between $y = 4x$ and $y = x^2$	5	L52–54
3	Find the area between $y = x^2 - 4$ and the axis on $[0, 3]$	5	L52–54
4	Volume when $y = \sqrt{x}$ on $[0, 9]$ is rotated about Ox	4	L52–54
5	Estimate $\int_0^2 \sqrt{1 + x^3} \, dx$ with 4 strips, and say over or under	6	L55–57

WHERE THE MARKS HIDE

Q2 gives one mark for solving $4x = x^2$ to get the limits. Q3 gives two for splitting at $x = 2$. Q5 gives two for the over/under statement **with its reason**. Five of twenty-five are for the setting up, not the arithmetic.

The four errors

ERROR	LOOKS LIKE	KIND
no split at a zero	Q3 answered as one integral, giving -3	method
limits guessed	Q2 done on $[0, 1]$	method
$+ C$ in a definite answer	$"= 10 + C"$	knowledge
ordinates miscounted	4 strips with 4 values of y	careless



The limits drill

Eight pairs of curves on the board. For each, the class calls out only **the limits**, in five seconds:

CURVES	LIMITS
$y = x, y = x^2$	$0, 1$
$y = 4x, y = x^2$	$0, 4$
$y = 9 - x^2, y = 0$	$-3, 3$
$y = x^2 - 4, y = 0$	$-2, 2$
$y = x^3, y = x$	$-1, 0, 1$

THE LAST ROW HAS THREE

$x^3 = x$ at $-1, 0, 1$, and the curves swap over at 0 . Two integrals are needed, or the parts will cancel.

02 Worked examples

EXAMPLE 1 Model answer, Q3: area between $y = x^2 - 4$ and the axis on $[0, 3]$.

- ① Zero at $x = 2$ inside the interval.

Split. Two marks.

② $\int_0^2 = \frac{8}{3} - 8 = -\frac{16}{3}$

③ $\int_2^3 = (9 - 12) - (-\frac{16}{3}) = \frac{7}{3}$

④ $\frac{16}{3} + \frac{7}{3} = \frac{23}{3}$

ANSWER

$$\frac{23}{3} \approx 7.67$$

EXAMPLE 2 Model answer, Q2: area between $y = 4x$ and $y = x^2$.

① $4x = x^2 \Rightarrow x = 0, 4$

One mark.

② The line is above between them.

③ $\int_0^4 (4x - x^2) dx = [2x^2 - \frac{x^3}{3}]$

④ $= 32 - \frac{64}{3} = \frac{32}{3}$

ANSWER

$$\frac{32}{3} \approx 10.67$$

EXAMPLE 3 A learner answered Q3 as -3 . Name the mistake.

① They integrated straight through the zero.

② The parts below and above the axis cancelled.

A method error.

③ Correct: split at $x = 2$.

ANSWER

Method error — no split at the zero

03 Interactive model

Run the limits drill before the rewrite; it is the mark most often lost.

Limits, splits and signs

Limits for $y = 4x$ and $y = x^2$:

$0, 1$

$0, 4$

$1, 4$

$-4, 4$

The area between them:

$\frac{4}{3}$

$\frac{16}{3}$

$\frac{32}{3}$

16

$y = x^2 - 4$ on $[0, 3]$ must be split at:

0

2

3

nowhere

That area is:

Volume from $y = \sqrt{x}$ on $[0, 9]$:

4 strips need how many ordinates?

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Limits of integration	Integrallash chegaralari	Пределы интегрирования
Area between curves	Egri chiziqlar orasidagi yuza	Площадь между кривыми
Volume of revolution	Aylanish jismi hajmi	Объём тела вращения
Trapezium rule	Trapetsiya usuli	Метод трапеций
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Sketch	Chizma	Эскиз

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The limits of an area question come from:

the question always

solving for the intersections

the axis

guessing

A curve crossing the axis inside the interval needs:

nothing

a split at the zero

squaring

doubling

A definite integral's answer includes:

$+ C$

just a number

a function

a family

The trapezium estimate for a concave-up curve is:

exact

too large

too small

unpredictable

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\int_1^4 2x \, dx$

ANSWER 15

2. $\int_0^2 3x^2 \, dx$

ANSWER 8

3. $\int_1^9 \frac{1}{\sqrt{x}} \, dx$

ANSWER 4

4. Limits for $y = 4x, y = x^2$

ANSWER 0, 4

5. Split point for $y = x^2 - 4$ on $[0, 3]$

ANSWER $x = 2$

6. Volume from $y = \sqrt{x}$ on $[0, 9]$

ANSWER $\frac{81\pi}{2}$

7. 4 strips need how many ordinates?

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. $\int_1^4 (x^2 - 2x) dx$

ANSWER 6

2. Area between $y = 4x$ and $y = x^2$

ANSWER $\frac{32}{3}$

3. Area between $y = x^2 - 4$ and the axis on $[0, 3]$

ANSWER $\frac{23}{3}$

4. Estimate $\int_0^2 \sqrt{1 + x^3} dx$ with 4 strips

ANSWER ≈ 3.28

5. Over or under, and why?

ANSWER over — concave up

6. Area between $y = x$ and $y = x^3$ on $[0, 1]$

ANSWER $\frac{1}{4}$

7. Volume from $y = 2x$ on $[0, 3]$

ANSWER 12π

Hard · stretch and reason

7 PROBLEMS

1. Area enclosed by $y = x^3$ and $y = x$

ANSWER $\frac{1}{2}$

2. Area between $y = x^2$ and $y = 2 - x^2$

ANSWER $\frac{8}{3}$

3. Volume from $y = \frac{1}{x}$ on $[1, 4]$ about Ox

ANSWER $\frac{3\pi}{4}$

4. Find k with area between $y = kx$ and $y = x^2$ equal to $\frac{9}{2}$

ANSWER $k = 3$

5. Estimate $\int_1^3 \frac{1}{x} dx$ with 8 strips

ANSWER ≈ 1.1032

6. Show the exact value is $\ln 3$

ANSWER The antiderivative of $\frac{1}{x}$

7. Explain why splitting matters in Q3

ANSWER Otherwise the two signed parts cancel

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. Find the limits before writing any integral.

1. Rewrite in full every question that lost a mark, showing how the limits were found.
2. Five problems from the section your knowledge column was heaviest in.

3. Find the area between $y = 6x$ and $y = x^2$, and between $y = x^2 - 9$ and the axis on $[0, 4]$.
4. Estimate $\int_0^1 \sqrt{1 + x^4} \, dx$ with 4 strips and state whether it is an over- or an under-estimate.